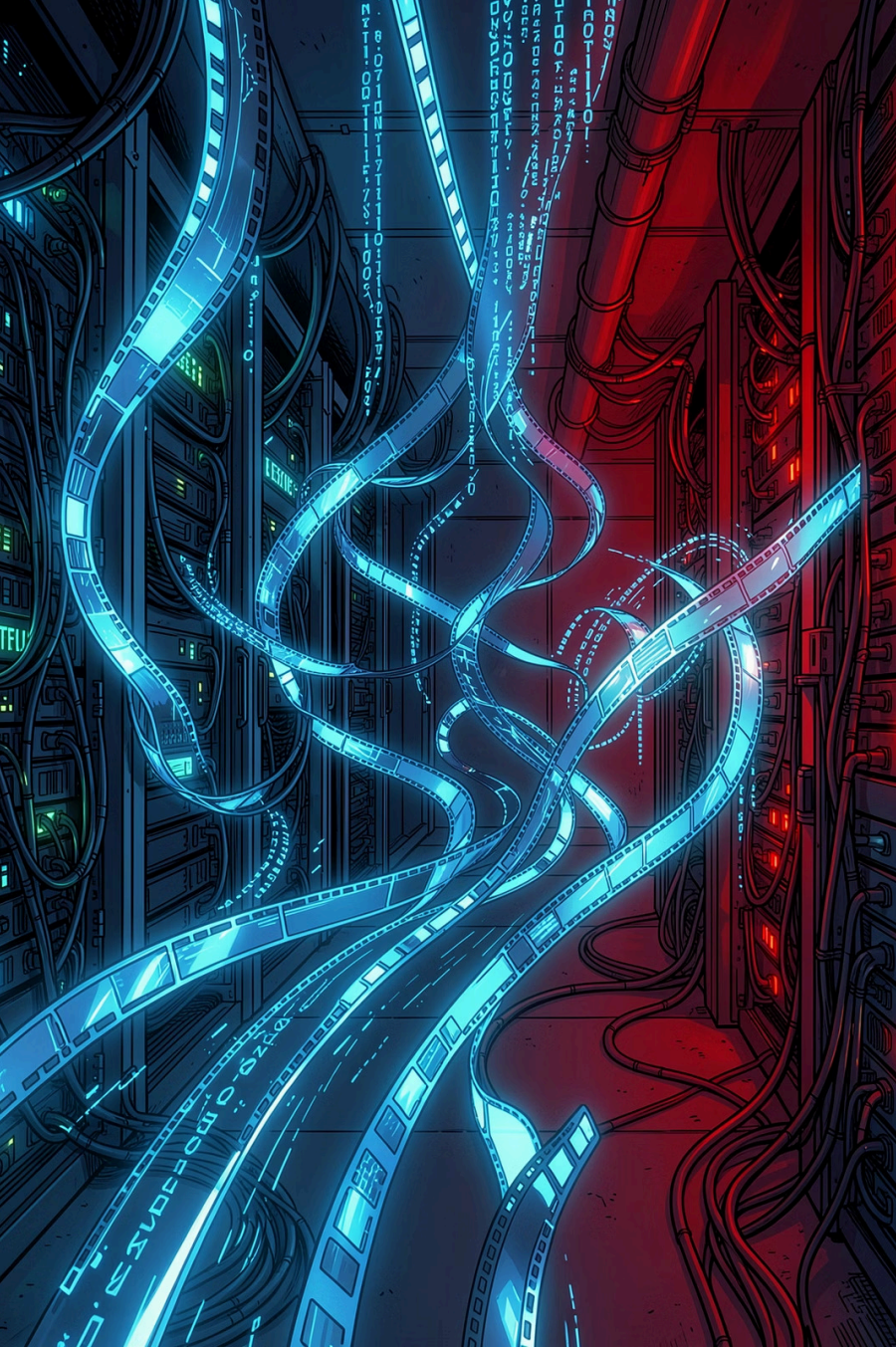




PERSONALISED RECOMMENDATION SYSTEM

AMRUTHA R D (24113016)

RAJNATH PRAJAPATI(24113104)

The Challenge of Sparsity

100M+

Ratings Analyzed

From 480K users across 17,770
movies

98.84%

Matrix Sparsity

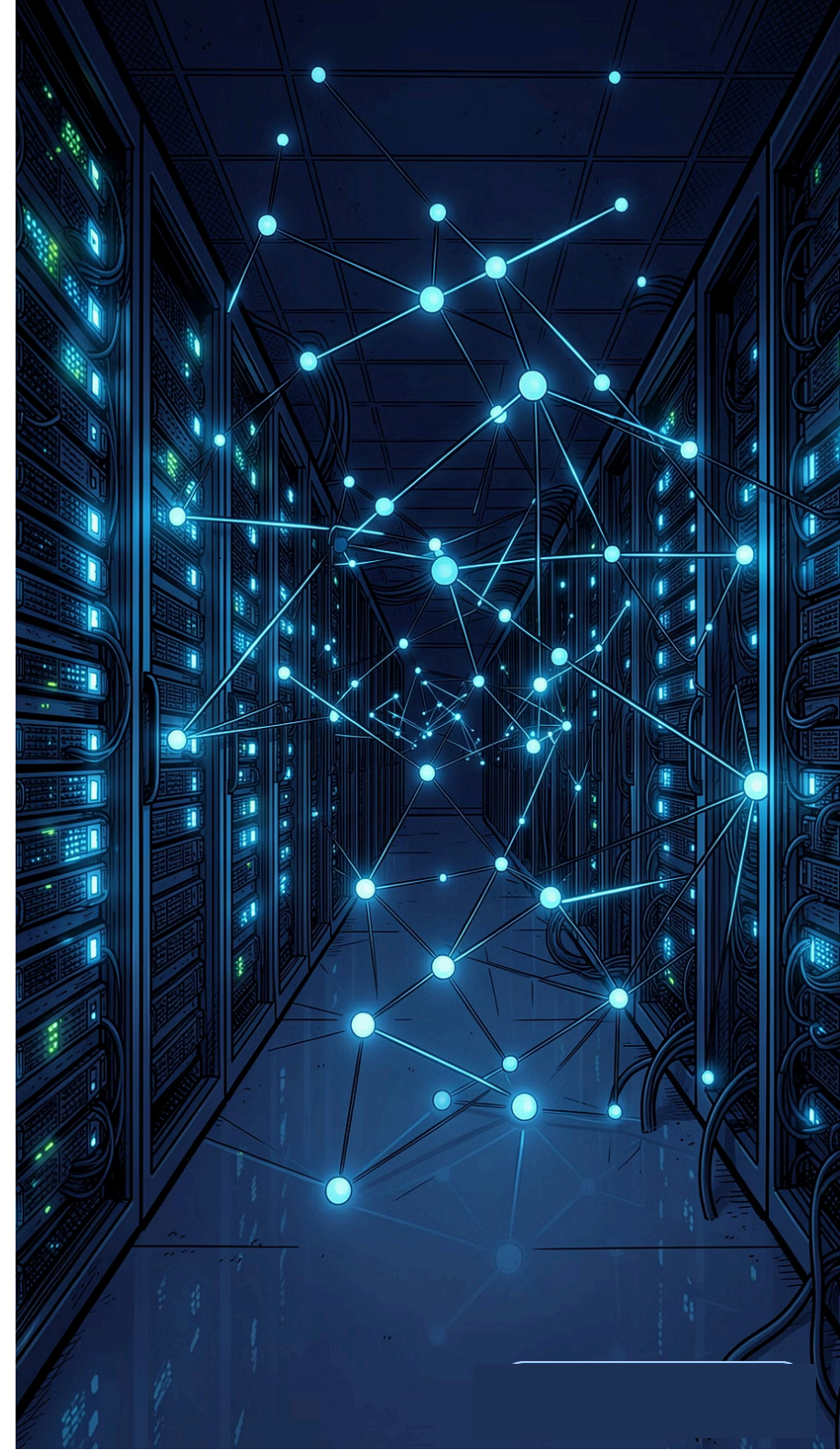
Standard collaborative filtering is
computationally non-viable at this
scale

17,770

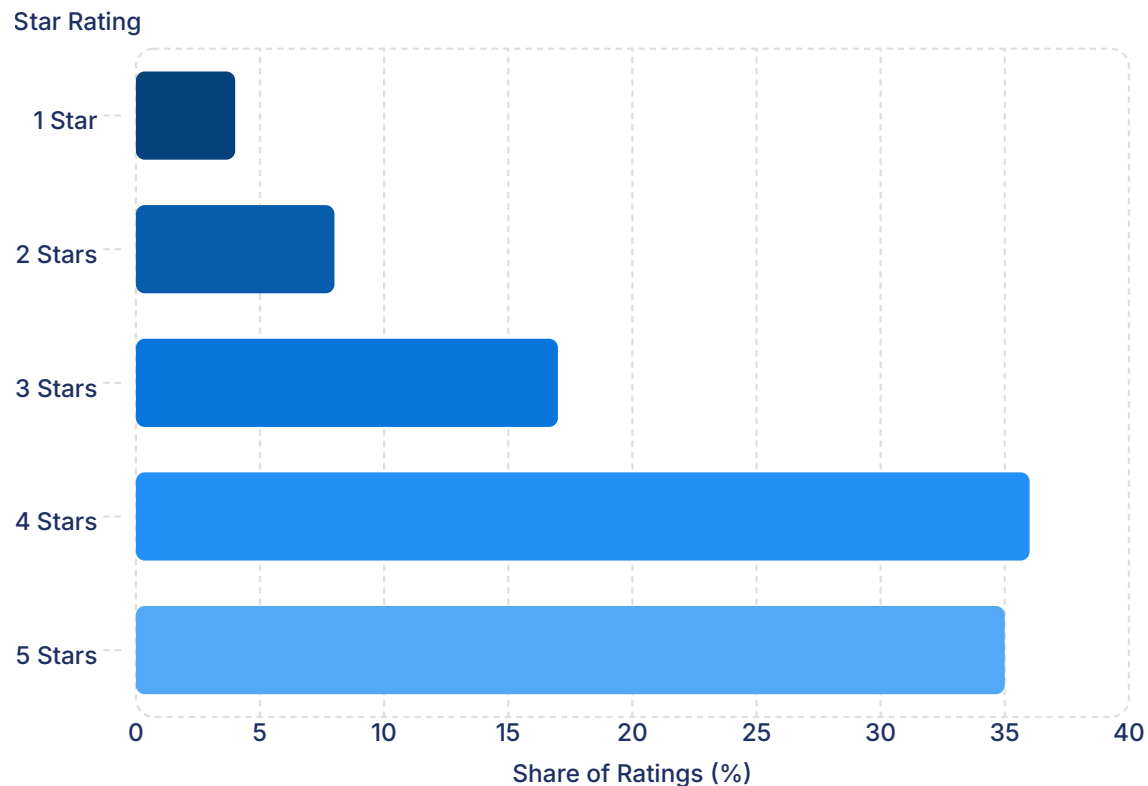
Unique Titles

Thousands of niche items sit
unobserved in the long tail

Objective: Generalize from extremely limited observations to predict unseen ratings and generate high-utility Top-K recommendation lists.



Key Insights: Exploratory Data Analysis



Selection Bias

Ratings exhibit a massive **positivity skew** — 4 and 5-star ratings dominate the corpus. 1-star reviews account for less than 5% of all observations.

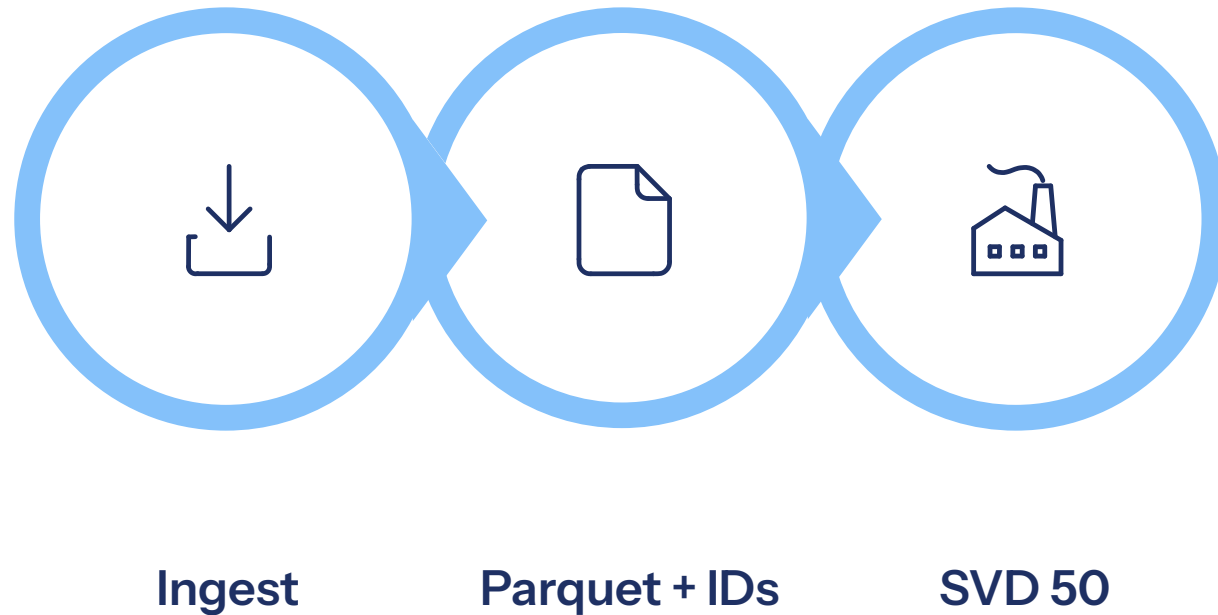
User Behavior

Activity follows the **Pareto Principle**: a tiny fraction of "Power Users" contribute the majority of training signal.

Cold Start Constraint

The **Long Tail** reality necessitates a shift from neighborhood-based methods to latent factor modeling.

Technical Approach: Data Pipeline & Baseline



The pipeline transforms raw rating logs into an optimized columnar format, enabling rapid iteration during model training and evaluation.

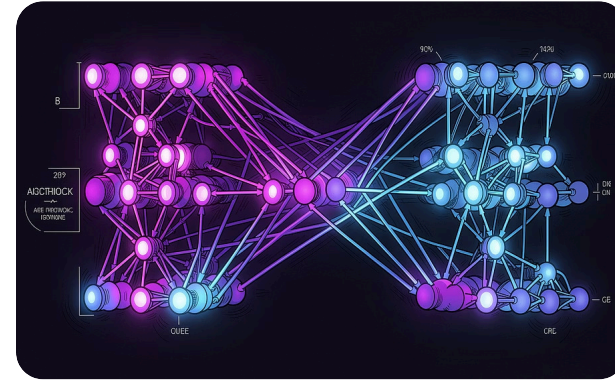
Optimization

Raw text converted to **Parquet format**, delivering a 10× improvement in I/O throughput during training epochs.

Model 1: Truncated SVD

A fast linear algebra baseline decomposing the user-item matrix into **50 latent factors** to capture global taste patterns. Serves as the performance floor for all subsequent models.

Captures **Preference Drift** by introducing time-varying user biases b_{ut} across **30 time bins**. Optimized with vectorized mini-batch SGD to model how user tastes evolve over months and years.



A **PyTorch-based** deep learning approach. Replaces the linear dot-product with a **Multi-Layer Perceptron (MLP)** to learn complex, non-linear interactions between user and item embeddings.

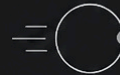
Experimental Results: The Metric Scorecard

Accuracy Metrics

Temporal-MF excels at predicting exact star ratings, achieving the lowest RMSE and MAE by modeling temporal dynamics in user preferences.

Ranking Metrics

Neural-CF consistently outperforms on **MAP@10**, **NDCG@10**, and **Hit Rate@10** — the metrics that matter most for content discovery and list quality.

	 SVD	 Temporal-MF	 Neural-CF
 Inference Speed	 fastest inference	 moderate ranking	 moderate ranking
 RMSE/MAE Accuracy	 baseline RMSE/MAE	 best RMSE/MAE accuracy	 moderate ranking
 Ranking Quality	 moderate ranking	 moderate ranking	 best MAP@10/NDCG@10/ HitRate@10
 Catalog Coverage	 moderate coverage	 moderate coverage	 strongest catalog coverage
 Compute Cost	 low cost	 moderate cost	 highest compute cost

Recommendation Examples: Personalized Discovery

Success Case — User 185251

Top-10 list accurately identifies a niche affinity for "**Sci-Fi with intellectual depth**" — surfacing *Inception* and *Interstellar* with high confidence.

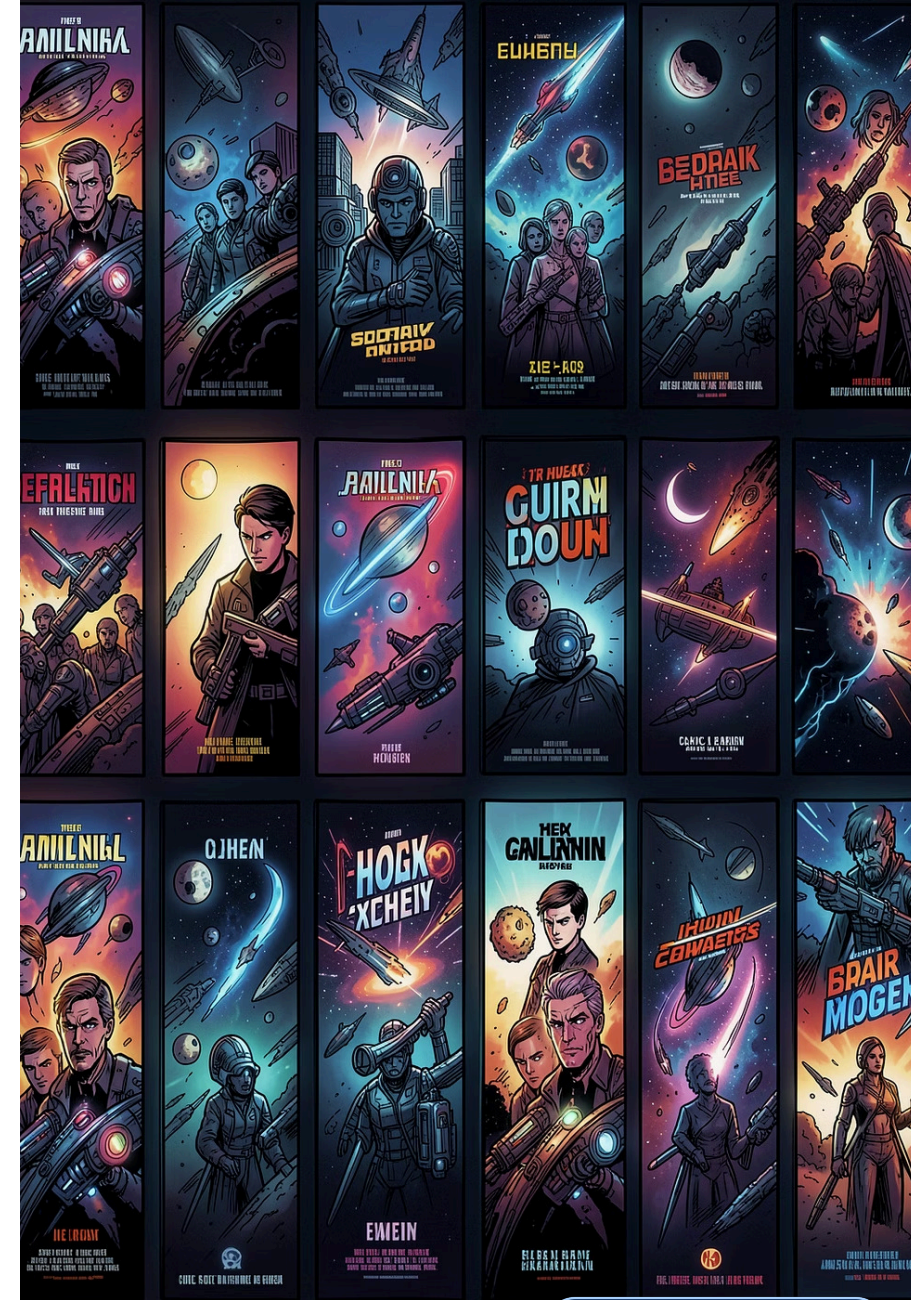
Latent Space Similarities

Item-item similarities reveal learned structure: *The Matrix* is mathematically linked to *Star Wars* in embedding space, validating semantic clustering.

Probe Set Validation

Hit Rate accuracy confirmed against the held-out **Probe validation set**, demonstrating generalization beyond the training distribution.

alxvc



Conclusion & Practical Usability

Model Trade-offs

SVD

Speed & simplicity

Temporal-MF

Rating accuracy

Neural-CF

Ranking quality

Cold Start Strategy

- **Global Popularity fallback** for new users with no rating history
- **Metadata injection** (release year, genre) for new items

Next Steps

- Expand Neural MLP to ingest **multi-modal data** (genres, directors, cast)
 - Build a true **Hybrid Recommender** combining collaborative and content-based signals
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